# Mingyun Kang

Researcher in Construction Management & Automation

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## Education

**Seoul National University of Science and Technology** *Seoul, Republic of Korea*
*Bachelor in Architectural Engineering, School of Architecture* *Mar 2017 – Feb 2024*
- Double Major: Business Administration
- GPA : 3.8 / 4.5

**Seoul National University of Science and Technology** *Seoul, Republic of Korea*
*Master in Architectural Engineering (Integrated Bachelor-Master Program)* *Mar 2024 – Aug 2025*
- GPA : 4.22 /4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System (Advisor : Prof. Taehoon Kim)

## Research Experience

**Research Assistant** *Seoul, Republic of Korea*
*CONICT Lab, Seoul National University of Science and Technology* *Dec 2021 – Aug 2025*
Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites.
- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

**Post-Master's Researcher** *Goyang, Republic of Korea*
*Korea Institute of Civil Engineering and Building Technology (KICT)* *Nov 2025 – present*
Post-Construction Evaluation & Management Center, Conducted research on Post-Evaluation of SOC in Republic of Korea

## Publications (SCIE)

**Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System** Dec 2024
Applied Sciences, 15(1), 125.
**Mingyun Kang**, Sebeen Yoon, Taehoon Kim
10.3390/app15010125 (Applied Sciences)

**Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model** Apr 2025
Applied Sciences, 15(9), 4875.
Juho Han, Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
10.3390/app15094875 (Applied Sciences)

**Automated Vision-based Location Monitoring for Jack Support** Jan 2025
KSCE Journal of Civil Engineering, 100374.
Sebeen Yoon, **Mingyun Kang**, Minhyung Kim, Taehoon Kim
10.1016/j.kscej.2025.100374 (KSCE Journal of Civil Engineering)

## Publications (SCOPUS)

**Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets** June 2024
ISARC Vol. 41, pp. 1176-1181. IAARC Publications.
**Mingyun Kang**, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim
10.22260/ISARC2024/0152 (ISARC 2024 — International Symposium on Automation and Robotics in Construction)

**Comparative Study of Structure from Motion on Construction Site** July 2025
ISARC Vol. 42, pp. 1395-1400. IAARC Publications.
**Mingyun Kang**, Sangmin Lee, Sebeen Yoon, Taehoon Kim
(ISARC 2025 — International Symposium on Automation and Robotics in Construction)

## Publications (International Conference)

**Computer Vision-Based Tile Counting Model for Automated Progress Monitoring** June 2023
CCC 2023, Keszthely, Hungary.
**Mingyun Kang**, Wooseok Lee, Junsang Yoo, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2023))

**A Preliminary Study on Automatic Jack Support Installation Interval Measurement Model** June 2023
CCC 2023, Keszthely, Hungary.
Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
(Creative Construction Conference (CCC 2023))

**Approach for Improving Panoramic Image Segmentation Quality on Construction Sites using SAM** June 2024
CCC 2024, Prague, Czech Republic.
Juho Han, Sanghyeon Na, **Mingyun Kang**, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2024))

## Publications (Domestic)

**Computer Vision-based Adhesion Quality Inspection Automation Model for Exterior Insulation and Finishing System** Mar 2023
JKIBC, 23(2), 165-173. (KCI)
Sebeen Yoon, **Mingyun Kang**, Hyeonseung Jang, Taehoon Kim
(Journal of the Korean Institute of Building Construction (JKIBC))

**Preliminary Study on Semantic Segmentation-based Quality Inspection Model for Exterior Insulation** Oct 2022
AIK, 42(2), 704-705.
Sebeen Yoon, **Mingyun Kang**, Byeongmin Lee, Taehoon Kim
(Proceedings of the Architectural Institute of Korea)

**A Basic Study on Image-based Numerical Measurement Model for Construction Sites** May 2023
KIBC Spring 2023, 23(1), 287-288. [Best Paper Award]
Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
(Proceedings of KIBC Spring Conference)

**Applicability Enhancement of Deep Learning-based Computer Vision Models for 360° Panoramic Images** Nov 2023
KICEM 2023, Goseong, 205-206.
**Mingyun Kang**, Junsang Yoo, Wooseok Lee, Taehoon Kim
(Proceedings of KICEM Annual Conference)

**A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information** May 2024
KIBC Spring 2024, Gunsan, 243-244.
Cheolhun Park, Jieun Lee, Jihyeon Lee, Yunjae Jeon, **Mingyun Kang**, Taehoon Kim
(Proceedings of KIBC Spring Conference)

**Performance Comparison Study on Image-based 3D Space Generation** Oct 2024
AIK Autumn 2024, Gyeongju, 974-975.
**Mingyun Kang**, Sebeen Yoon, Taehoon Kim
(Proceedings of Architectural Institute of Korea Autumn Conference)

**A Basic Study on 3D Reconstruction Performance Improvement Using 360° Camera** Nov 2024
KIBC Autumn 2024, 203-204.
**Mingyun Kang**, Juho Han, Sebeen Yoon, Taehoon Kim
(Proceedings of KIBC Autumn Conference)

**Construction of High-Resolution 360° Image Dataset Using Image Super-Resolution Technology** Nov 2024
National Student Conference 2024, 177-178.
Minhyung Lee, **Mingyun Kang**, Taehoon Kim
(Proceedings of National University Student Academic Conference)

**A Basic Study on Image Analysis-based Crackdown on Illegal Rooftop Buildings** May 2025
KIBC Spring 2025, Gyeongju, 309-310.
Yeon Choi, Seonwoo Lee, Yujin Oh, Gangmin Bae, **Mingyun Kang**, Taehoon Kim
(Proceedings of KIBC Spring Conference)

## Patents

**Automatic Supervision Method and Device for Adhesive Application Quality** Apr 2023
Domestic Patent Application No. 10-2023-0053756. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**

**Automated Construction Site Supervision System** Sept 2023
Domestic Patent Application No. 10-2023-0125464. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**

**Tile Quantity Counting System** Oct 2023
Domestic Patent Application No. 10-2023-0135013. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**, Wooseok Lee, Junsang Yoo

**Method for Measuring Rebar Spacing Based on Single Image Information** Sept 2024
Domestic Patent Application No. 10-2024-0127210. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, **Mingyun Kang**, Cheolhun Park, Jihyeon Lee, Jieun Lee, Yunjae Jeon

**Method for Detecting Illegal Rooftop Buildings Based on Video Image Analysis** Aug 2025
Domestic Patent Application No. 10-2025-0117608. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, **Mingyun Kang**, Seonwoo Lee, Yeon Choi, Yujin Oh, Gangmin Bae

## Projects

**Digital Mapping-based Platform Research Laboratory for Healthy Construction Lifecycle** *June 2022 – Feb 2025*

National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

- Vision-based adhesive application quality inspection for EIFS
- Tile quantity counting via computer vision

**Development of Digital-based Architectural Supervision and Construction Automation Robot Technology** *Apr 2022 – Dec 2026*

Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

- Instance segmentation of insulation finishing systems using synthetic datasets
- Panoramic image segmentation for indoor construction sites

**Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites** *Apr 2025 – Dec 2027*

Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

- XR-based human-robot collaboration interface
- Integrated robot operation system for construction sites

## Awards

**Best Paper Award (최우수학술상)** May 2023

Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement Model for Construction Sites

Korean Institute of Building Construction (KIBC)

**Best Paper Award (최우수학술상)** May 2024

Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information

Korean Institute of Building Construction (KIBC)

**Excellence Paper Award (우수학술상)** May 2025

Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops

Korean Institute of Building Construction (KIBC)

**Outstanding Thesis Award (우수논문상)** Aug 2025

Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System

Seoul National University of Science and Technology

## Languages & Skills

- Korean (Native) · English (Advanced) · Python (Advanced)

## Hobbies

- Golf · Running · Snowboarding

## Military Service

**Sergeant (병장), Rifleman** *Republic of Korea*

*Capital Defense Command, 1st Security Battalion, Republic of Korea Army* *Dec 2018 – Sept 2021*

Completed mandatory military service as an infantry rifleman. Honorably discharged at the rank of Sergeant.

## Certificates

**Architectural Engineer License (건축기사)** *June 2025*

**TOEIC Speaking — Advanced Low (Score 160)** *Nov 2023*

## References

**Professor Taehoon Kim**